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SUPERSEDING
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MILITARY SPECIFICATION

AIR CONDITIONER, A/M32C-18 TRAILER MOUNTED, DIESEL ENGINE DRIVEN

This specification is approved for use by the Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense

1. SCOPE

1.1 Scope. This specification covers one type of diesel-engine-driven, trailer-mounted air conditioner. Air conditioners procured under this specification shall be warranted by the manufacturer as specified in the procurement documents.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

F-F-300	Filter, Air Conditioning, Viscous-Impingement and Dry Types, Cleanable
BB-F-1421	Fluorocarbon Refrigerants
QQ-B-654	Brazing Alloy, Silver
RR-C-271	Chains and Attachments, Welded, Weldless, and Roller Chain
VV-F-800	Fuel Oil, Diesel
VV-L-825	Lubricating Oil, Refrigerant Compressor
WW-T-799	Tube, Copper, Seamless, Water and Refrigeration (For use with Solder-, Flared- or Compression-Type Fittings)

MILITARY

MIL-P-514	Plate, Identification, Instruction and Marking, Blank
MIL-R-3593	Refrigeration and Cooling Equipment (excluding Household Refrigerators), Packaging of
MIL-E-5220	Extinguisher, Fire, Aircraft, Type A-20
MIL-T-5624	Turbine Fuel, Aviation, Grades JP-4 and JP-5

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: ASD/ENESS, Wright-Patterson AFB, OH 45433 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-M-8090	Mobility, Towed Aerospace Ground Equipment, General Requirements for
MIL-A-8421	Air Transportability Requirements, General Specification for
MIL-P-23377	Primer Coating, Epoxy-Polyamide, Chemical and Solvent Resistant
MIL-D-38386	Duct Assembly, Ground, Conditioned Air, Insulated, Flexible
MIL-T-83133	Turbine Fuel, Aviation, Kerosene Type, Grade JP-8
MIL-C-83286	Coating, Urethane, Aliphatic Isocyanate, for Aerospace Applications

STANDARDS

FEDERAL

FED-STD-595	Colors

MILITARY

DOD-STD-100	Engineering Drawing Practices
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of US Military Property
MIL-STD-143	Standards and Specifications, Order of Precedence for the Selection of
MIL-STD-281	Automobiles, Trucks, Truck-Tractors, Trailers and Trailer Dollies, Preservation and Packaging of
MIL-STD-461	Electromagnetic Interference Characteristics Requirements for Equipment
MIL-STD-462	Electromagnetic Interference Characteristics, Measurement of
MIL-STD-808	Finishes, Protective, and Codes, for Finishing Schemes for Ground and Ground Support Equipment
MIL-STD-810	Environmental Test Methods
MIL-STD-831	Test Reports, Preparation of
MIL-STD-882	System Safety Program for Systems and Associated Subsystems and Equipment: Requirements for
MIL-STD-889	Dissimilar Metals
MS33562	Connection, Aircraft Ground Air Conditioning, 8-Inch, Minimum Requirements
MS51336	Lunette Coupler, Drawbar, Rings

DRAWINGS

AIR FORCE

64C1004-2	Connector-Air Duct Hose, Ground Cooling, Male
64C1005-2	Connector, Air Duct Hose, Ground Cooling, Female
64C1006-2	Packing Preformed - Air Duct Hose, Ground Cooling
7545352	Requirements for Finishes, Protective, and Codes for San Antonio ALC Ground and Ground Support Equipment

US ARMY MOBILITY EQUIPMENT RESEARCH AND DEVELOPMENT CENTER

13216E6168	Valve, Charging with Cap

(Code Identification Number 97403)

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PUBLICATIONS

AFSC DESIGN HANDBOOK

1-11 Air Transportability

(Copies of specification, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

ENVIRONMENTAL PROTECTION AGENCY

Control of Air Pollution from New Motor Vehicles and New Motor Vehicle Engines

(Application for copies should reference "Code of Federal Regulations 45CFR Part 1201" and should be addressed to the Superintendent of Documents, US Government Printing Office, Washington, DC 20402.)

AMERICAN SOCIETY OF HEATING, REFRIGERATING, AND AIR CONDITIONING ENGINEERS, INCORPORATED

ASHRAE Standard 15 USA Standard Safety Code for Mechanical Refrigeration
ASHRAE Standard 37 Methods of Rating and Testing Unitary Air Conditioners
ASHRAE Guide and Data Book

(Application for copies should be addressed to the American Society of Heating, Refrigeration, and Air Conditioning Engineers, Incorporated; 345 East 47th Street; New York, NY 10017.)

AIR CONDITIONING AND REFRIGERATION INSTITUTE

ARI Standard 210 Standard for Unitary Air Conditioning Equipment

(Application for copies should be addressed to Air Conditioning and Refrigeration Institute; 1815 North Fort Myer Drive; Arlington, VA 22209.)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS

ASME Boiler and Pressure Vessel Code
Code VIII Pressure Vessels
ASME Power Test Code 19.5:4
Supplements on Instruments and Apparatus
Part 2, Pressure Measurements; Chapters 1, 2, and 4
Part 3, Temperature Measurement; Chapter 6
Part 4, Head Measuring Apparatus
Part 5, Measurements of Quantity of Materials

(Application for copies should be addressed to the American Society of Mechanical Engineers, United Engineering Center, 345 East 47th Street, New York, NY 10017.)

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AMERICAN NATIONAL STANDARDS INSTITUTE, INC.

S1.4-1971 Sound Level Meters, Specification for

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018.)

NATIONAL FIRE PROTECTION ASSOCIATION

NFPA No. 70 National Electrical Code (Electric Wiring and Apparatus)

(Application for copies should be addressed to the National Fire Protection Association; 470 Atlantic Avenue; Boston, MA 02210.)

SOCIETY OF AUTOMOTIVE ENGINEERS, INC. (SAE)

J 537 Storage Batteries

(Application for copies should be addressed to the Society of Automotive Engineers; 400 Commonwealth Dr.; New York, NY 10001.)

TIRE AND RIM ASSOCIATION, INC.

Tire and Rim Association Inc., Yearbook

(Application for copies should be addressed to the Tire and Rim Association, Inc.; 3200 West Market Street; Akron, Ohio 44131.)

3. REQUIREMENTS

3.1 Product conformance testing. When specified (see 6.2), the contractor shall furnish sample unit(s) for product conformance inspection and approval (see 4.4 and 6.2b).

3.2 Standard air conditioner and accessories. The air conditioner components, assemblies, and accessories shall meet or exceed the requirements of this specification.

3.3 Components. The Air conditioner shall consist of the following major components or systems:

Description	See Requirements
Trailer chassis	3.8
Refrigeration system	3.9
Air supply system	3.10
Fuel system	3.11
Controls and instrumentation	3.12
Enclosure	3.13.

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3.4 Selection of standards and specifications. Standards and specifications for necessary commodities not identified in the manufacturer's source codes shall be selected in accordance with MIL-STD-143 or as specified herein.

3.5 Materials.

3.5.1 Metals. Metals shall be of the corrosion-resistant type or a type treated to resist corrosion due to fuels, salt fog, or atmospheric conditions likely to be met in storage or normal service.

3.5.1.1 Dissimilar metals. Unless protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals are defined in MIL-STD-889.

3.5.2 Gasketing and insulating materials. Where gasketing and insulating materials are used, they shall be compounded to insure their oil resistance and their ability to withstand the environmental conditions specified herein.

3.6 Design and construction. The air conditioner shall be so constructed that no parts will work loose in service. It shall be built to withstand the stresses, jars, vibrations, and other conditions incident to shipping, storage, towing, and operational service.

3.6.1 Reliability. A 435-hour specified mean-time-between-failure (MTBF) at a confidence level of 0.9 shall be established by performing a 900-hour endurance test with no failures, to demonstrate compliance with the specified MTBF.

3.6.2 Maintainability. Air conditioners furnished under this specification shall have the following maintainability features:

a. Minimum number of parts consistent with the performance specified herein.

b. Minimum amount of training and time necessary for assembly, disassembly, location of trouble sources, and maintenance including servicing. Where practicable, parts and components shall be located or positioned for rapid and simple inspection and recognition of excessive wear or potential failure.

c. Permit adjustments, servicing, replacement of parts and components, and other maintenance with minimum disturbance to other parts or components. Parts and components shall be located for ample and rapid access. In performing maintenance, if engineering reasoning or data determines that physical or visual interference between items cannot be avoided, the item predicted to require the most maintenance shall be located for best accessibility.

d. Permit maintenance with general purpose tools. The use of any special purpose tools and equipment shall be approved by the procuring activity.

e. Minimum number of tools required for maintenance by design practices such as reducing the variety of bolthead sizes to the practicable minimum.

3.6.3 Exposed items (see 6.3.4). All connectors, junction boxes, and other exposed items shall be completely weatherproof.

3.6.4 Safety features.

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3.6.4.1 Safety of personnel. Provisions for the safety of personnel shall be incorporated to the maximum extent possible with regard to anticipated operating conditions and the capability of operating personnel. The danger to personnel from fire and explosion shall be avoided by separation of hazardous substances from heating sources, incorporation of vents and drains, and other fire prevention measures. The air conditioner shall conform to the recommendations of the USA Standard Safety Code for Mechanical Refrigeration (ASHRAE Standard 15) and the paragraph of MIL-STD-882 titled, System safety requirements.

3.6.4.2 Fire extinguisher. A Type A-20 1-quart fire extinguisher in accordance with MIL-E-5220 shall be provided.

3.6.4.3 Hazardous materials. Brazing alloys containing silver shall not be used in any location where they are likely to come in contact with solutions containing ethylene glycol.

3.6.4.4 Guards. Guards shall be provided on the air conditioner to shield rotating or hazardous components. The guards shall be removable to permit maintenance and servicing.

3.6.4.5 Belt drives. All belt drives used in the air conditioner shall utilize belts which are oil resistant and conduct static electricity.

3.6.5 Air transportability. The air conditioner shall be air transportable in accordance with MIL-A-8421 and AFSC Design Handbook 1-11 in types C-130, C-141, and C-5A aircraft.

3.6.5.1 Tiedowns. The air conditioner shall be provided with ten tiedown points located such that the vehicle can be restrained in all directions during air transport. Tiedown points shall be designed for vehicle restraint against the forces specified in MIL-A-8421.

3.6.5.2 Clearance. Sufficient clearance shall be maintained around all tiedown points to allow for restraining chains without interference from any other structure on the air conditioner. The tiedown points shall have sufficient clearance to permit attachment of a chain tiedown with a grabhook.

3.6.5.3 Ramp clearance. The air conditioner shall be capable of negotiating a 17° ramp in both forward and reverse directions.

3.6.5.4 Fuel tank. The air conditioner fuel tank shall have sufficient strength and rigidity to meet the air transportability requirements of MIL-A-8421, para 3.3.4 when filled to three-quarters (3/4) capacity with fuel.

3.7 Performance.

3.7.1 Ambient operating range. The air conditioner shall be capable of starting and operating continuously in the cooling mode and furnishing the required cooling capacity within the range of ambient conditions specified

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by the crosshatched area of figure 1. Further, the air conditioner shall be capable of operating in the heating and ventilating modes under the following ambient conditions.

a. Heating mode: -53.9 to 23.9°C (-65 to 75°F)

b. Ventilating mode: -53.9 to 51.7°C (-65 to 125°F).

3.7.1.1 Low ambient temperature. The necessary components and accessories shall be included to permit starting and continuous operation of the air conditioner in the heating and ventilating modes at ambient temperatures down to -53.9°C (-65°F).

3.7.2 Basic rated cooling output. While operating at sea level in an ambient temperature of 37.8°C (100°F) dry bulb (DB) and 29.4°C (85°F) wet bulb (WB), the air conditioner shall be capable of continuously delivering 450 pounds per minute (ppm) of airflow at 10.0°C (50°F), 95 percent maximum relative humidity (RH), and 35 inches water gage (wg) static air pressure.

3.7.3 Maximum rated cooling output. The air conditioner shall be capable of continuously delivering 450 ppm of air flow at 10.0°C (50°F), 95 percent maximum RH, and 35 inches wg static air pressure while operating at sea level in an ambient of 51.7°C (125°F) DB and 23.9°C (75°F) WB temperatures.

3.7.4 Selected cooling output. The air conditioner shall be capable of continuously delivering outlet air temperatures ranging from 4.4°C (40°F) to 21.7°C (70°F) at airflow rates ranging from 20 to 100 percent of the rated airflow. The discharge air temperature shall be automatically controlled after the desired conditioned air temperature is manually selected by the operator. The static discharge air pressure shall be selectively adjustable from 0 to 35 inches wg static air pressure.

3.7.5 Continuous cooling output. The air conditioner shall be capable of continuously providing 450 ppm of airflow at 10.0°C (50°F), 95 percent maximum RH, and 35 inches wg static air pressure while operating at a barometric pressure of 24.0 inches mercury (Hg). The ambient temperature shall be 37.8°C (100°F) DB, and 17.2°C (63°F) WB.

3.7.6 Basic rated heating output. While operating in an ambient temperature of -31.7°C (-25°F) the air conditioner shall be capable of delivering 600,000 BTU/hr of heating capacity. The airflow rate shall not exceed 400 ppm while delivering at 35 inches wg static air pressure.

3.7.6.1 Low temperature heating output. While operating in an ambient temperature of -53.9°C (-65°F), the air conditioner shall be capable of delivering 600,000 BTU/hr of heating capacity. The airflow rate shall not exceed 450 ppm while delivering 35 inches wg static air pressure.

3.7.7 Selected heating output. The air conditioner shall be capable of continuously delivering heated air at temperatures up to 65.6°C (150°F). The discharge air temperature shall be automatically controlled after the temperature is manually selected by the operator. The airflow rate shall be fully adjustable ranging from 20 to 100 percent of the 450 ppm rated airflow. The static discharge air pressure shall be selectively adjustable from 0 to 35 inches wg static air pressure.

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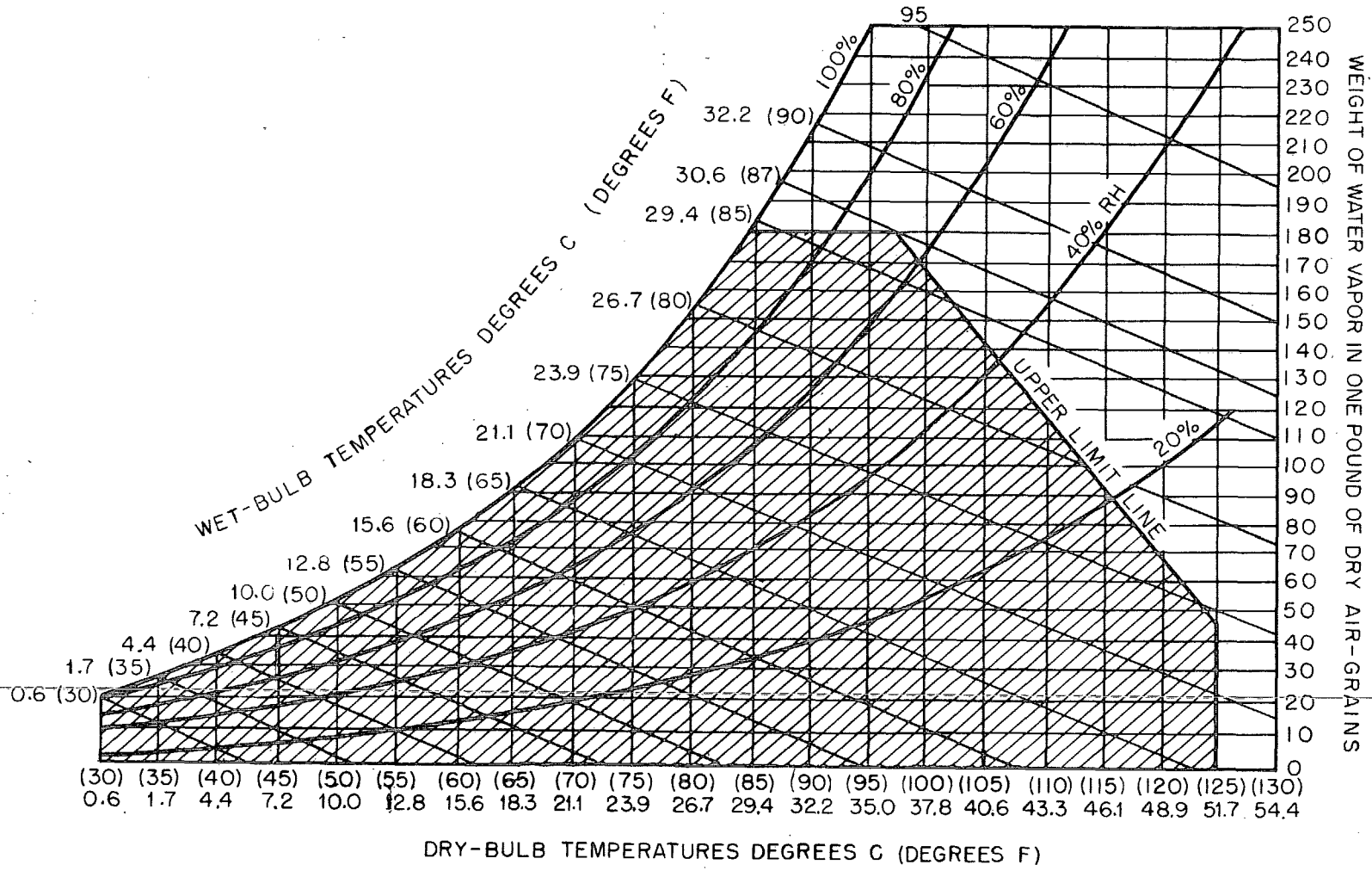


FIGURE 1. Ambient range for cooling operation.

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3.7.8 Tilted position. The air conditioner shall operate satisfactorily while tilted in any plane at an 8-1/2° angle from the true horizontal.

3.7.9 Environmental conditions. The air conditioner shall operate satisfactorily under the following conditions.

a. Operating temperatures ranging from -53.9° to 51.7°C (-65° to 125°F).

b. Relative humidity up to 100 percent, including conditions wherein condensation takes place in the form of water and frost.

c. Fungus growth as encountered in tropical and subtropical regions.

d. Salt atmosphere as encountered near bodies of salt water and in transportation by ship.

e. Sand and dust as encountered in desert locations.

f. Rainfall as encountered in any locale.

g. Solar radiation as encountered in desert locations.

h. Vibration incident to service use.

i. Operation at altitudes from sea level to 6,000 feet.

3.7.9.1 Other environmental conditions. The air conditioner shall satisfactorily withstand exposure (non-operating) to the following conditions:

a. Storage temperatures from -53.9°C to 71.1°C (-65° to 160°F).

b. Pressures ranging from 30 to 3.4 inches Hg, approximately an altitude of 50,000 feet.

c. Shock forces anticipated in transport.

3.7.10 Operating fuels. The air conditioner shall be capable of meeting the performance requirements specified herein while using DF-2 in accordance with VV-F-800. The engine shall also be operable using DF-A in accordance with VV-F-800, JP-4 and JP-5 in accordance with MIL-T-5624 and JP-8 in accordance with MIL-T-83133.

3.7.11 Rotating speeds. Air conditioner rotating components shall not exceed the maximum speed recommended by the component manufacturer for continuous operation.

3.8 Trailer chassis.

3.8.1 Chassis. The chassis shall comply with the requirements specified herein.

3.8.2 Diesel engine drive. The air conditioner shall have one diesel engine of proven performance. The engine shall be capable of continuously furnishing all required horsepower to permit satisfactory air conditioner operation between the ambient extremes specified herein. The engine shall be required to

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start and operate satisfactorily under all environmental conditions specified herein. The diesel engine shall have a continuous horsepower rating to meet all the performance requirements specified herein.

3.8.2.1 Engine filters. The engine shall have an oil filter, primary and secondary inlet fuel filters, and an engine inlet air filter recommended for use with the engine by the engine manufacturer. The air filter shall have a readily-visible pressure drop indicator to indicate engine inlet air filter condition.

3.8.2.2 Exhaust system. An engine exhaust pipe shall be provided to carry engine exhaust gas away from the conditioned air inlet. The exhaust gases shall be discharged vertically.

3.8.2.2.1 Engine muffler and spark arrestor. The engine exhaust system shall include a muffler and spark arrester that will effectively reduce engine exhaust noise to within the sound level limitations specified in table I. The muffler and spark arrestor shall be of a size that will prevent creation of excessive backpressure on the engine and a type that will preclude the discharge of sparks from the exhaust stack.

TABLE I. Noise level.

Measurement Position	Maximum permissible A-weighted sound level in dB (Ref dB: 20 $\mu N/m^2$)
Operator's position	96
30-foot radius	92
60-foot radius	88

3.8.2.2.2 Engine exhaust covers. The engine exhaust stack shall have a hinged cover which is easily opened by the exhaust gas flow of the engine during startup and normal operation. The cover shall be plated or protected in a manner to inhibit formation of corrosive deposits.

3.8.2.3 Heavy duty engine cooling system. The air conditioner engine cooling system shall include a metal surge tank. A sight gage shall be provided in the surge tank. The sight gage shall be located in a position readily visible for checking coolant level. The cooling system furnished shall maintain the coolant temperature below the boiling point.

3.8.2.4 Clutches and flexible couplings. The refrigerant compressor shall be driven by the engine through a flexible coupling. A manually-operated clutch will be provided to insure satisfactory engine startup and operation at the low temperatures specified herein. The clutch actuator shall be readily accessible but need not be located on the control panel.

3.8.2.5 Fuel storage tank. The chassis shall include an integral fuel tank. The fuel tank shall be capable of retaining all fuel, without spillage, during all testing specified herein.

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3.8.2.6 Component alignment. Alignment pins shall be used for satisfactory component alignment to insure proper operation and performance of the air conditioner.

3.8.2.7 Mobility. The air conditioner shall have type II, group C mobility in accordance with MIL-M-8090, with the exceptions and additions specified herein.

3.8.2.7.1 Towing surfaces. The surfaces on which the air conditioner is to be towed shall be limited to level paved highway, graded gravel road, and test course for type I, class 2 vehicles. The air conditioner shall be towable at a speed of 20 miles per hour (mph) on paved highway, 10 mph on graded gravel roads and 5 mph on the test course for type I, class 2.

3.8.2.7.2 Brakes.

3.8.2.7.2.1 Service brakes. The air conditioner shall have service brakes which are automatically activated by deceleration of the towing vehicle.

3.8.2.7.2.2 Parking brakes. The air conditioner shall have parking brakes. The parking brakes shall be capable of holding the air conditioner on a 17° slope whether headed in the forward or reverse direction.

3.8.2.7.3 Tires and wheels. The tires and wheels shall be in accordance with the Tire and Rim Association, Inc., Yearbook. They shall meet the mobility requirements specified herein. Tire valves shall be readily accessible for tire inflation pressure testing. Tires and wheels shall be easily removed.

3.8.2.7.4 Towbar and lunette eye. The lunette eye shall be in accordance with MS 51336. A torsion spring shall be provided on the towbar of the air conditioner which assists in lifting and/or holding the towbar while connecting or disconnecting from a tow vehicle. The torsion spring shall not be used to transmit towing forces from the towing vehicle to the air conditioner. The spring shall be designed to withstand the normal stresses caused by towbar operation. The lifting force needed to lift the towbar shall be 5 to 25 pounds when the air conditioner is resting on a level surface. Further, the spring shall be designed to reduce the amount of force which it imparts to the towbar as the towbar is raised to the full upright position. The towbar shall be designed to operate satisfactorily from -53.9°C (-65°F) to 51.7°C (125°F). The measurement of the specific lifting force shall be made at 23.9°C (75°F).

3.8.2.7.5 Steering. The maximum cramping angle permitted by the steerable front wheels shall not result in damage to the rear axle assembly. The inside cramping angle shall not be less than 40°.

3.9 Refrigeration system.

3.9.1 Compressor. The compressor shall be of the reciprocating type and of sufficient capacity to meet the requirements specified herein. The compressor shall incorporate automatically-operated cylinder unloaders for starting and operating at reduced refrigeration loads. The compressor shall be capable of continuous operation at refrigerant pressures encountered during operation in ambient temperatures to 51.7°C (125°F).

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3.9.1.1 Compressor drive. The compressor shall be engine driven. The compressor shall be of sufficient capacity to provide the performance specified herein without exceeding the compressor manufacturer's maximum continuous speed.

3.9.1.2 Lubrication system. The compressor shall be lubricated by means of an integral lubrication system. The lubricant shall be of a type conforming to VV-L-825. An oil reservoir of sufficient capacity shall be provided for proper lubrication of the compressor under all conditions. A readily visible oil-level sight gage shall be provided in the compressor crankcase.

3.9.2 Refrigerant system.

3.9.2.1 Refrigerant. The refrigerant system shall be designed for use with refrigerant conforming to BB-F-1421, monochlorodifluoromethane (refrigerant type 22).

3.9.2.2 Sight glass. A sight glass shall be provided in the liquid refrigerant line.

3.9.2.3 Pressure vessels. The receiver, shall be in accordance with Section VIII of the ASME Boiler and Pressure Vessel Code.

3.9.2.4 Refrigerant piping. All refrigerant piping shall be type L copper tubing conforming to WW-T-799. Only high-temperature solder conforming to QQ-B-654 shall be used for assembly of refrigerant piping, except where such use might result in a safety hazard. The suction line shall be sized to provide sufficient refrigerant flow velocities to insure return of oil to the compressor.

3.9.2.4.1 Vibration eliminators. The refrigerant piping system shall include vibration eliminators in both suction and discharge lines of the compressor. The vibration eliminators shall be satisfactory for the pressures to be encountered as well as for use with the refrigerant.

3.9.2.5 Pumpdown control. A control shall be provided for automatic pumpdown of the refrigerant system prior to engine shutdown. This shall be accomplished by the use of a solenoid shutoff valve located in the liquid refrigerant line between the receiver and expansion valve. Subsequent to shutoff of this valve, the compressor shall operate until the compressor suction pressure is reduced to a value that will actuate the low-pressure cutoff switch. A solenoid valve or check valve shall be located in the discharge line of the compressor. The valve shall isolate the refrigerant charge subsequent to the pumpdown operation. It shall be possible to operate the engine in the start, idle, and run ranges without these operations being affected by the pumpdown control.

3.9.2.6 Refrigerant receiver. An ASME refrigerant receiver of adequate size to store liquid refrigerant after pumpdown shall be provided. An automatic pressure relief valve shall be provided on the receiver. The relief valve shall be selected to permit satisfactory operation and storage of the air conditioner at all temperatures specified herein.

3.9.2.6.1 Refrigerant quantity indicator. The receiver shall have a liquid refrigerant quantity indicator. The indicator shall be graduated in increments noting each 1/4 of full level.

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3.9.2.7 Servicing valves. The refrigerant system shall incorporate both a discharge and suction side servicing valve. The servicing valves shall comply with ERDL Drawing 13216E6168. Sufficient servicing valves shall be provided to permit complete evacuation of the refrigerant system.

3.9.2.8 Check valves. A check valve and a solenoid valve shall be provided in the refrigerant system to prevent refrigerant migration into the compressor lubrication oil.

3.9.2.9 Filter-drier. A replaceable, cartridge type, filter-drier shall be provided. The filter-drier shall be located between the receiver and the thermostatic expansion valve of the refrigerant system. The filter-drier shall be sufficiently isolated by a solenoid and a manual shut-off valve to preclude unnecessary loss of refrigerant during filter-drier replacement.

3.9.3 Evaporator assembly. The evaporator assembly shall be the extended-surface type having copper tubes with aluminum fins. The coil shall be capable of withstanding, without permanent deformation or leakage, an internal proof pressure of 500 psig tested underwater. Air velocity entering the coil shall not exceed 500 fpm. The external portions of the evaporator enclosure shall be insulated after painting.

3.9.3.1 Frost prevention. Means shall be provided to preclude frost from accumulating on the evaporator coils while the air conditioner provides continuous operation without a decrease in the cooling output.

3.9.3.2 Expansion valve. The flow of refrigerant to the evaporator coil shall be controlled by a thermostatic-type expansion valve. The valve shall be of the pressure limiting, externally-equalized, and fully-adjustable type.

3.9.3.2.1 Strainer. The expansion valve shall be equipped with an inlet strainer.

3.9.3.3 Evaporator condensate drain. A drain pan shall be provided and the drain shall be properly sized and located so that all water falling into the pan will be ejected through the drain line to the outside of the air conditioner when the air conditioner is tilted at an 8-1/2° angle in any direction. The drain pan shall be sealed to prevent the dripping of condensate on internal components.

3.9.3.4 Refrigerant distribution. A venturi-type refrigerant distributor shall be provided to distribute the flow of refrigerant through the evaporator coil. The distribution lines shall be of equal length.

3.9.4 Condenser.

3.9.4.1 Condenser coil. The condenser shall be of an air-cooled, extended-surface type having copper tubes with aluminum fins. Condensation from the evaporator section shall not be used to provide additional cooling effect. The condenser shall have adequate surface area to enable the air conditioner to perform its cooling function under the extreme ambient conditions specified herein. The condenser shall withstand, without permanent deformation or leakage, an internal proof pressure of 500 psig, tested underwater.

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3.9.4.2 Condenser fans. The condenser coils shall be cooled by fans of adequate capacity for removing condenser heat. The fans shall be statically balanced.

3.10 Air supply system.

3.10.1 Conditioned air blower. A blower shall be provided to supply conditioned air and shall meet the airflow and pressure requirements specified in 3.7.2 through 3.7.5. The blower shall not overload the prime mover. The blower shall be dynamically balanced for the required operating speeds.

3.10.2 Airflow regulation. The air conditioner shall have individually-operated manual airflow controls on each conditioned air outlet to maintain the airflow rates specified in 3.7.2.

3.10.2.1 Internal ducts. Insulation shall be provided where necessary to meet performance requirements specified herein.

3.10.2.2 Dump valve. The conditioned air supply plenum shall have a manually-operated dump valve to relieve the conditioned air pressure inside the conditioned air ducting. The dump valve control shall be located at the control panel of the air conditioner. Operation of the dump valve shall relieve the conditioned air pressure such that the thrust (reactionary) force due to pressure forces and momentum change acting on the 12-inch to 8-inch adapters does not exceed 10 pounds when the dump valve and the three 12-inch outlets are fully open. Normal opening of the dump valve under any of the ambient conditions specified herein shall not cause the conditioned air blower motor or other system(s)/component(s) to create conditions which cause failures, safety devices to actuate, or unsafe operation of the air conditioner including that due to dump valve discharge direction.

3.10.3 Conditioned air outlet connectors. The air conditioner shall have three 12-inch conditioned air outlet connectors conforming to AF Drawing 64C1004-2. Each connector shall be furnished with a quick-disconnect cover which effectively seals the connector when used with the preformed packing described by AF Drawing 64C1006-2. Each connector cover shall be of the captive style and fastened to the air conditioner by a suitable length of weldless steel chain in accordance with RR-C-271 for type II, Class 6 chain. The connectors shall be located side by side with sufficient clearance to permit quick connection of the conditioned air ducts. The air conditioner shall have a hinged quick-access door to provide easy access to the conditioned air outlet connectors.

3.10.4 Conditioned air duct assemblies. The air conditioner shall be provided with six, 25-foot length, insulated flexible air ducts. The ducts shall be of 12-inch nominal inside diameter in accordance with MIL-D-38386 (with connectors). The connectors and preformed packing shall be in accordance with AF Drawing 64C1004-2, 64C1005-2, and 64C1006-2, respectively. An air duct assembly having a female connector shall readily connect to either of the three male connectors to be provided on the air conditioner.

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3.10.5 Conditioned air duct connector adapters. Three 12-inch to 8-inch duct adapters shall be provided with each air conditioner. The 12-inch end of the connector shall mate with a connector conforming to AF Drawing 64C1004-2. The 8-inch end of the duct adapter shall mate with an MS33562 aircraft connector. The duct adapter shall incorporate a tapered 12-inch to 8-inch transition section and shall contain a quick-disconnect device to unhook the duct adapter from the MS33562 aircraft connector. Each duct adapter shall not exceed 15 pounds in weight. The air conditioner shall be operable in each mode while the adapters are stored internally.

3.10.6 Duct storage. The six flexible duct assemblies and the three conditioned air duct adapters shall be stored internally. The air conditioner shall be capable of operation in each mode while the ducts are stored in the enclosure.

3.10.6.1 Storage racks. Duct storage racks shall be provided for the flexible duct assemblies and the three conditioned air duct adapters specified herein.

3.10.7 Screens. To prevent the entrance of large foreign objects, the condenser coil and engine air-intake openings in the air conditioner shall be protected by screens such that a 5/8-inch diameter rod will not pass through.

3.10.7.1 Air supply system filters. A cleanable-type filter shall be provided on the inlet to the conditioned air blower. The filter shall comply with the requirements of F-F-300.

3.11 Fuel system.

3.11.1 Fuel tank. The air conditioner shall have an internal fuel tank of sufficient capacity to permit 6 hours of continuous operation at rated cooling output. A vent for the fuel tank shall be provided.

3.11.1.1 Fill opening. The fuel tank shall be equipped with an opening of not less than 3-inches inside diameter. The fill opening shall be located on an external wall. The fill opening shall be within easy reach while standing on ground level. A red fill opening cap shall be provided. The fill opening cap shall be of a captive style.

3.11.1.1.1 Fill opening strainer. A removable screen shall be provided in the fill opening. The screen shall be 80 mesh with .0055 inch diameter wire.

3.11.1.2 Fuel tank drain. A fuel tank drain shall be provided to permit complete draining of the fuel tank. The drain shall be readily accessible and shall discharge all fuel directly to the ground without contacting other air conditioner components.

3.11.2 Fuel lines. All fuel lines in the air conditioner shall be constructed of steel or other fire-resistant material. Where flexibility is required in these lines, flexible, fire-resistant coupled hose assemblies shall be used. Connections in metal fuel lines shall not employ materials that will be affected by temperatures up to 676.7°C (1,250°F).

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3.11.3 Sealing provision. All gaskets, synthetic rubber seals, et cetera, shall be suitable for use with the fuels specified in 3.7.10, both at the temperatures likely to be encountered prior to actuation of a device to signal overheating of the air conditioner and at the cold ambient temperatures specified herein.

3.11.4 Draining provisions. The air conditioner shall have adequate provisions for draining from the compartments any fuel leakage from fuel lines or controls. All fuel lines shall be easily disconnected and removed for servicing.

3.11.5 Fuel gage. A fuel quantity gage shall be provided. The gage shall be located on the control panel of the air conditioner. The gage shall be graduated in 1/4 increments of full fuel level.

3.12 Controls and instrumentation.

3.12.1 Operating controls. The air conditioner shall have adequate controls for starting and controlling the engine drive and controlling the temperature and airflow rates. After startup and initial adjustment of the operating controls, the air conditioner shall be capable of operating without an operator in attendance. All controls and instruments shall be located on the control panel unless otherwise specified herein.

3.12.1.1 Governor control. A variable speed governor shall be provided. The governor shall permit adjustment of engine speed above idle speed by manual adjustment of the governor within the operating speed range of the air conditioner.

3.12.1.2 Throttle control. The engine shall be governed when the throttle is in the open position. The throttle shall not interfere with governor operation. The throttle control shall permit the engine to be operated at normal idle speeds.

3.12.1.3 Ignition switch. An ignition switch shall be provided on the air conditioner to start and stop the engine.

3.12.1.4 Conditioned air temperature control. Controls shall be provided so that the cooling capacity can be manually adjusted, as required, to obtain conditioned air temperatures ranging between 4.4° and 21.1°C (40° and 70°F) at the airflow rates specified herein. Controls shall also be provided so that the heating capacity can be manually adjusted at the control panel to obtain conditioned air temperatures between -6.7° and 65.6°C (20° and 150°F) at airflow rates ranging from 20 to 100 percent of the airflow rates specified in 3.7.2 and static pressures ranging from 0 to 35 inches wg.

3.12.1.5 Conditioned airflow control. Each conditioned air outlet shall have an independent airflow control to permit individual control of the airflow rates specified herein.

3.12.1.6 Mode switch. The air conditioner shall have a mode selector switch which indicates whether the air conditioner is in the cooling, heating, pumpdown/airdump or ventilate mode of operation.

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3.12.1.7 Emergency stop switch. A stop switch shall be provided on the air conditioner which immediately stops operation of the unit under all operating modes. The unit shall not go through the automatic pumpdown cycle when this switch is actuated. The emergency stop switch must stop engine intake airflow in addition to other applicable operations.

3.12.1.8 Circuit breakers. The circuit breakers shall be properly rated for use under all performance and environmental conditions specified herein.

3.12.2 Safety controls. The safety controls shall be either a manual or automatic reset, provided the air conditioner ceases operation when a safety control is actuated. As a minimum, the following safety controls shall be provided:

a. Cut off the engine when there is a loss of engine oil pressure.

b. Cut off the engine when the engine coolant temperature becomes too low.

c. Cut off the engine when the fuel pressure is too low.

d. Cut off the engine when the engine speed exceeds allowable limits.

e. Cut off the engine when refrigerant suction pressure becomes too low.

f. Cut off the engine when refrigerant discharge pressure becomes too high.

g. Cut off the engine when the compressor oil pressure drops too low.

h. Pressure relief valve on the receiver to protect the refrigerant system from abnormally high pressures.

i. Circuit breakers for electrical control circuits and electrical motors to protect against overload conditions.

j. High temperature cutoff switches for the heating system to protect against over-temperature conditions.

k. An air filter switch to cut off the blower motor if the filters are not installed.

l. An air pressure switch to preclude energizing the electric heaters unless the blower is operating.

m. Control panel door safety switch to prevent operation when the control panel door is not properly secured.

3.12.3 Instrumentation.

3.12.3.1 Instrument and control panel. The air conditioner shall be provided with a composite instrument and control panel upon which all instruments, gages, hourmeters, throttle levers, switches, and warning lights for the refrigeration and air supply system are to be mounted. The control panel shall be readily accessible to operating personnel, and all items on the panel shall be permanently and legibly marked. The panel shall be illuminated by a manually-controlled light. Ready access to the rear of the control panel shall be provided.

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3.12.3.1.1 Instruments. Gauges and instruments shall be color coded with normal operating ranges and/or limitations. Instruments providing a direct indication of the following shall be among those included on the instrument panel:

a. Conditioned air temperature °C (°F).

b. Engine drive speed revolutions per minute (rpm).

c. Engine coolant temperature.

d. Engine oil pressure.

e. Blower motor current (amps).

f. Fuel gage.

g. Airflow gages (pounds per minute).

h. High compressor discharge pressure indicator.

i. Low compressor suction pressure indicator.

j. Low compressor oil pressure indicator.

k. Conditioned air static pressure gages (inches of water gage).

3.13 Enclosure. The refrigeration and conditioned air supply system shall be fully enclosed in a metal enclosure. All openings shall be covered by suitable panels, doors, or screens.

3.13.1 Weather resistance. The enclosure shall be constructed to resist weather conditions and to prevent rain, sand, and dust from collecting within the enclosure in quantities that would be detrimental for proper operation of the air conditioner. Doors or panels which must be open during operation of the air conditioner shall be inspected and tested to insure that components and accessories will operate properly while exposed to the environmental conditions specified herein.

3.13.2 Doors. All exterior openings that must be opened during operation shall be provided with access doors or screens. In addition, all working parts of the air conditioner shall be readily accessible through convenient weathertight doors for maintenance and servicing operations. All doors or panels shall be equipped with latches to hold them securely in the closed position. Doors hinged at the side shall be equipped with fixtures that will hold the doors firmly open.

3.13.3 Safety chains. All caps and covers, except the fuel cap, which are removed during operation or servicing of the air conditioner shall be retained by a safety chain. The safety chains shall be steel, conforming to type II, class 6 chain under RR-C-271. The length of the chains will be consistent with the operating or servicing requirements of the air conditioner.

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3.13.4 Fluids and lubricants. Unless otherwise specified, (see 6.2), each air conditioner shall be delivered with required quantities of fluids and lubricants, except fuel, necessary for proper operation. The coolant system shall contain sufficient ethylene-glycol solution to provide freeze protection in an ambient temperature of -53.9°C (-65°F). Lubricants and fluids which must be changed to permit low temperature operation shall be clearly identified by an instruction plate.

3.14 Electrical system.

3.14.1 Operating voltage. The operating voltage of the engine starting system shall be 24 volts direct current (VDC). Batteries shall be provided that have an adequate amp-hour rating to permit startup and operation of the air conditioner under the environmental extremes specified herein. The batteries shall be retained in a battery box inside the enclosure of the air conditioner and shall be readily accessible to maintenance personnel. Battery rating shall be as stated in SAE 537.

3.14.2 Electrical circuits. The air conditioner electrical circuits shall contain insulated electrical wiring capable of withstanding the operating and environmental conditions specified herein. All electrical wiring shall use copper conductors and shall have visible numerical coding for ready identification by maintenance personnel. Electrical wiring used in the air conditioner shall not exceed the recommended capacity rating stated in the National Electrical Code.

3.14.2.1 Overcurrent protection. Circuit breakers shall be provided in control circuits, primary branch circuits, and electric motor circuits to protect electrical components from damage due to overcurrent conditions.

3.14.2.2 Overload protection. Electric motors shall be protected from damage due to thermal overload. The overload protection devices to be used shall be current sensing, temperature operated overload relays.

3.14.2.3 Grounding. The air conditioner electrical circuits shall be provided with a common ground. .

3.14.2.4 Wiring and cables. All wiring and cables used in the electrical circuits shall be firmly supported to prevent undue stress on the conductor terminals and eliminate the possibility of a shift in conductor position. Wiring shall be protected against wear due to abrasion. Wires passing through holes in sheet metal shall be protected by rubber grommets or cable clamps. Clearance between wiring and heat-generating parts shall be sufficient to prevent damage to wire insulation. Sufficient slack shall be provided to prevent stress on cable forms, wires, and terminal connections, and to facilitate replacement of parts without disconnecting other parts.

3.14.2.5 Terminations. Before being soldered to terminals, wires shall be effectively secured to the terminals so that the connections are not dependent on soldering alone for mechanical strength.

3.14.2.6 Terminals. Terminals attached to wires may be of either the solder or solderless type.

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3.14.2.7 Relays. All relays shall hold settings for long periods of use. The relays shall be capable of operating under the service conditions specified herein without readjusting the contact spring tension adjustment. Overload relays shall reset either automatically or have an external manual reset.

3.14.2.7.1 Relay coils. Except for relays that are hermetically sealed, the coils shall be impregnated to resist corrosion and accumulation of moisture.

3.14.2.7.2 Relay covers. Where practicable, relays shall be covered for mechanical protection and electrical shielding to minimize electromagnetic interference. Covers shall be rugged, properly fitted, weatherproof, and securely fastened to prevent loosening under the service conditions specified herein.

3.15 Part number of interchangeable parts. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The item identification and part number requirements of DOD-STD-100 will be applied to the manufacturer's part numbers and changes thereto.

3.16 Electromagnetic interference. The air conditioner, including chassis, shall meet the requirements specified in MIL-STD-461 for class III C equipment with the following exceptions:

a. The broadband radiated emission reading (with correction) shall not exceed 112 dBuV/m/MHz at a frequency of 150 KHz.

b. The broadband radiated emission reading (with correction) shall not exceed 105 dBuV/m/MHz at a frequency of 600 KHz.

c. The broadband radiated emission reading (with correction) shall not exceed 91 dBuV/m/MHz at a frequency of 9.6 MHz.

3.17 Acoustic noise level. When measured as specified in 4.5.5, the sound level generated by the air conditioner shall not exceed the values specified in table I.

3.18 Dimensions and weight.

3.18.1 Width. The overall width of the air conditioner shall not exceed 91 inches.

3.18.2 Height. The overall height of the air conditioner shall not exceed 104 inches.

3.18.3 Length. The overall length of the air conditioner shall not exceed 244 inches with the towbar in the upright and locked position.

3.18.4 Weight. The maximum overall weight of the air conditioner when equipped with the conditioned air ducts, duct adapters, and all fluids, except fuel, shall not exceed 18,100 pounds.

3.19 Color. The exterior finish of the air conditioner shall be semigloss green, Color No. 24052, in accordance with FED-STD-595.

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3.20 Finishes and protective coatings. Finishes and protective coatings shall be in accordance with MIL-STD-808 and AF Drawing 7545352 as specified for Type I exposure, except that primer paint films shall be in accordance with MIL-P-23377, and that the final paint films shall be in accordance with MIL-C-83286. Both interior and exterior surfaces shall be finished.

3.21 Operational markings.

3.21.1 Lettering and numbering. All lettering and numbering shall be performed using stencils as specified in MIL-STD-808. The letters and numbers applied using stencils will be at least 1/2 inch high and shall be applied directly over the exterior finish of the air conditioner.

3.21.1.1 Stenciled markings. The air conditioner shall have the following information stenciled on the exterior cabinet in the appropriate location.

a. Tire inflation pressure

b. Maximum towing speed - 20 mph

c. Approved fuels (DF-2 and JP-4)

d. Fire extinguisher location

e. Center of gravity

f. Battery location

g. Location of supply air outlets

h. Location of reducer couplings

i. Supply air duct storage

j. Warning to personnel not to enter unit while it is operating.

3.21.2 Data plates. Instruction and identification data shall be permanently and legibly etched on metal plates in accordance with MIL-P-514. The instruction plates shall be permanently attached in the most accessible and visible place possible.

3.21.2.1 Diagrams. Schematic diagrams of the refrigerant and electrical systems shall be provided. Each valve, switch, et cetera, on the diagram shall be properly identified.

3.21.2.2 Operating instructions. Brief instructions for starting and operating the heating, refrigeration, and air supply systems shall be permanently attached to the exterior of the enclosure on or near the control panel. The instruction plate shall list each step in the operation of the air conditioner.

3.21.2.3 Transportation data plate. A transportation data plate shall be mounted on the front of the air conditioner. The data plate shall contain at least the following:

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a. An isometric view of the air conditioner.

b. Shipping weight.

c. Loading cubage.

d. Overall height, width, and length.

3.21.2.4 Identification plates. Valves, switches, levers, and other controls used in the operation of the air conditioner shall be properly identified by identification plates. The identification plates shall indicate the relative position of the lever or control handle such as on, off, 24V, et cetera. The plates shall be permanently fastened and conspicuously located on or near the applicable controls or component.

3.21.2.5 Lubrication plate. A plate shall be permanently attached to the air conditioner in a conspicuous location on the unit. The plate shall state the type, grade or class of lubricant and the quantity required. In addition, the lubrication plate shall identify lubricants which must be changed to permit operation under extremely low ambient temperatures. The plate shall identify alternate lubricants to be used when the standard lubricant would be unsatisfactory.

3.21.2.6 Radio interference plate. An aluminum instruction plate containing the following information shall be installed in a conspicuous location on the air conditioner:

CAUTION

THIS AIR CONDITIONER IS EQUIPPED WITH ELECTROMAGNETIC INTERFERENCE SUPPRESSION DEVICES. CARE SHOULD BE EXERCISED DURING AIR CONDITIONER MAINTENANCE TO RETAIN AND REINSTALL THESE DEVICES PRIOR TO REUSE OF THE AIR CONDITIONER.

3.22 Identification of product. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130. A nameplate shall be securely attached to the air conditioner and shall contain the following information:

Air Conditioner A/M 32C-18

Manufacturer's name and part number

Rated performance:

Cooling - 66 tons at 450 pounds/minute

Heating 600,000 BTU/hr at -53.9°C (-65°F)

Refrigerant type R-22

NSN (assigned by procuring activity)

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Manufacturer's serial number

USAF Registration Number

Applicable contract number.

3.23 Workmanship. Welded, bolted, and riveted construction shall be in accordance with best commercial practice. Particular attention shall be given to alignment of parts, tightness of assembly screws and bolts, quality of exterior finish, refrigerant system solder joints, and legibility of operating controls and instruments.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary.

4.2 Classification of tests. The inspection and testing of the air conditioner shall be classified as follows:

Product conformance tests . . . . . See 4.4.3

Individual tests . . . . . See 4.4.4

Periodic sampling plan and tests . . . . . See 4.4.5

4.3 Test conditions.

4.3.1 Test fuels. Unless otherwise specified herein, the test fuels shall be grade DF-2 in accordance with VV-F-800 and JP-4 in accordance with MIL-T-5624. The engine manufacturer must certify that the engine will operate satifactorily on all fuels specified in 3.7.10.

4.3.2 Method of testing and rating. The method of testing and rating the air conditioner shall be in accordance with ASHRAE Standard 37. The air conditioner shall contain the proper quantity of all fluids and lubricants during testing under 4.4.3.

4.3.3 Apparatus.

4.3.3.1 Positive counter. Rotational speeds shall be determined by means of a directly coupled positive counter that will count the revolutions for not less than one minute. A combination unit that counts the revolutions while measuring the time required in connection with other data is preferred. Strobotachometers may be used if data can be furnished to attest to their accuracy. The use of hand tachometers shall not be acceptable.

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4.3.3.2 Nozzles or orifices. Airflow shall be measured by means of smooth-approach nozzles or flat-plate orifices conforming to ASHRAE Standard 37. Other apparatus such as pitot tubes and rotameters may be used when calibrated against recognized standards and approved by the procuring activity. The rate of airflow shall be expressed in terms of pounds per minute. Temperature shall be expressed in degrees Celsius (Fahrenheit).

4.3.3.3 Psychrometric chart. Humidity shall be determined by means of an ASHRAE psychrometric chart (ASHRAE Guide and Data Book) from readings of wet- and dry-bulb temperatures. Air velocity over the wet-bulb thermometer shall be in accordance with ASHRAE Standard 37. The thermometers shall be so arranged that the air passing over the wet bulb shall not subsequently pass over the dry bulb. The wick of the wet bulb shall be kept wet with distilled water only and shall be replaced whenever a perceptible quantity of oil, dirt, incrustation, or foreign substance has accumulated thereon.

4.3.3.4 Mercurial barometer. Barometric pressure shall be measured by means of a mercurial barometer at the same elevation as the test. The pressure shall be corrected for the temperature of the scale and Mercury and for variation in gravity resulting from the location of the barometer with regard to elevation and latitude. Readings obtained from aneroid barometers shall not be acceptable.

4.3.3.5 Thermocouples and potentiometers. Temperatures shall be measured by means of appropriately located thermocouples and properly calibrated potentiometers. When practicable, thermocouples shall be insulated from contact with other metals. Temperatures shall be expressed in degrees Celsius (Fahrenheit). Laboratory-type Mercury thermometers will be permissible if used correctly in the application.

4.3.3.6 Liquid manometers. Pressure and pressure differentials shall be measured by means of liquid manometers. However, pressures exceeding the range of conventional liquid manometers shall be measured by means of properly calibrated Bourdon-tube gages. Gages shall be of laboratory type having a dial diameter of not less than 4 inches.

4.3.4 Readings. Readings of all pertinent pressures, temperatures, rotational speeds, and other data necessary in demonstrating compliance with applicable paragraphs of Section 3 shall be recorded at intervals not exceeding 30 minutes, unless otherwise specified herein.

4.3.5 Fuel consumption. Fuel consumption measurements shall be made by either the volume or weight method. The quantity selected for the volume or weight method shall be such that each reading will cover an elapsed time of at least 1 minute. Flowmeter readings for fuel consumption shall be used only when the flowmeter and use thereof have been specifically approved by the procuring activity. All expression of quantities shall be recorded on a weight basis.

4.3.6 Air conditioner capacity. The discharge temperatures, airflow rates, relative humidities and static pressure specified herein shall be measured in the 8-inch sections simulating the three conditioned air duct adapters. All measurements shall be made in accordance with ASHRAE Standard 37 with a 25-foot section of flexible ducting attached to each conditioned air outlet. The ducting shall be in accordance with MIL-D-38386. (Capacity rating points are specified for standard sea level atmospheric pressure except where otherwise specified.)

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4.3.7 Preliminary run-in. Any preliminary run-in specified by the manufacturer shall be performed prior to submission of the air conditioner for testing. A certified log of all preliminary running time shall be included in the air conditioner log.

4.4 Product conformance testing

4.4.1 Test sample. The first production air conditioner shall be subjected to all tests specified in 4.4.3.

4.4.2 Test report. Upon completion of the product conformance tests, a report shall be prepared in accordance with MIL-STD-831.

4.4.3 Product conformance tests

Examination of product	See 4.5.1
Air transportability	See 4.5.2
Mobility	See 4.5.3
Electromagnetic interference	See 4.5.4
Acoustic noise level	See 4.5.5
Tilted position	See 4.5.6
Leak	See 4.5.7
Operational	See 4.5.8
Performance	See 4.5.9
Electrical	See 4.5.10
Duct storage	See 4.5.11
Environmental	See 4.5.12
Endurance	See 4.5.13
Manintainability demonstration	See 4.5.14
Final inspection	See 4.5.15.

4.4.4 Individual tests. Each air conditioner shall be subjected to the following tests as described under 4.5:

a. Examination of product	See 4.5.1
b. Leak tests	See 4.5.7
c. Operational tests	See 4.5.8
d. Electrical tests	See 4.5.10
e. Final inspection	See 4.5.15

4.4.5 Periodic sampling tests. One air conditioner shall be selected at random from the first 10 and one from each subsequent 10 or fraction thereof produced and subjected to the following tests:

Performance tests	See 4.5.9
Duct storage test	See 4.5.11

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4.4.5.1 Rejection and retest. When an item selected from a production run fails to meet the specification, items still on hand or later produced shall not be accepted until the extent and cause of failure have been determined and corrected. The contractor shall explain to the Government representative the cause of failure and the action taken to preclude recurrence. After correction, all of the tests shall be repeated.

4.4.5.1.1 Individual tests may continue. For production reasons, individual tests or other sampling plans may be continued pending the investigation of a sampling test failure. But final acceptance of the items on hand or produced later shall not be made until it is determined that the items meet all the requirements of the specification.

4.4.5.1.2 Defects in items already accepted. The investigation of a test failure could indicate that defects may exist in items already accepted. If so, the contractor shall advise the procuring activity of all the defects likely to be found and the method of correcting them.

4.5 Test methods.

4.5.1 Examination of product. The air conditioner shall be inspected for the following defects to determine compliance with the requirements specified herein:

a. Critical external dimensions not as specified

b. Parts or components missing or not as specified

c. Material not as specified

d. Total weight of the unit fully charged and ready to operate exceeds the weight specified

e. Parts hazardous to operating and maintenance personnel not fully enclosed, insulated, or properly guarded

f. Fans not statically balanced; blower wheel not dynamically balanced.

g. Marking and data plates not attached or not as specified

h. Fluids and lubricants not as specified or not at proper level

i. Panels not insulated

j. Evaporator pan or base leaking or not insulated

k. Component alignment not as specified

l. Undesirable control panel defects

m. Electrical wiring not as specified

n. Bolted and riveted connections not as specified

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o. Welding defects

p. Fan motors do not indicate proper direction of rotation

q. Direction of flow not indicated on filter-drier

r. Washers or lockwashers missing

s. Poor workmanship

t. Treatment and painting defects

4.5.2 Air transportability. Verify air transportability in accordance with MIL-A-8421 and AFSC DH 1-11. A dimensional analysis for loading and restraining the air conditioner in types C-130, C-141, and C-5A aircraft shall be provided.

4.5.2.1 Tire loading checks. The weight on each wheel of the air conditioner shall be measured with the air conditioner standing on a flat surface.

4.5.3 Mobility tests. The mobility tests shall be conducted on the first production air conditioner. Subsequent items shall not be accepted until mobility requirements have been satisfied and any necessary modifications have been accomplished. Each subsequent production item shall be constructed in accordance with the final approved configuration of the air conditioner. The air conditioner shall be subjected to an operational test in accordance with 4.5.8 after completion of the mobility tests.

4.5.3.1 Road tests. The air conditioner shall be towed not less than 300 miles over paved highway or similar improved roads and not less than 50 miles over graded unimproved roads. The air conditioner shall be towed for not less than 25 miles over a test course for type I, class 2 requirement. Towing speeds shall be as specified in 3.8.2.7.1. At least 10 short-radius right turns and 10 short-radius left turns shall be made. Throughout the road test, the air conditioner shall be observed as to its trackability, ease of handling, tendency to yaw, tendency to sway, etc. Upon completion of the road test, the air conditioner shall be inspected for excessive wear, misalignment, distortion, or any sort of structural failure or impending failure or malfunction of any part of the entire air conditioner. A complete description of any excessive wear, damage, failure, impending failure, or roadability deficiencies shall be recorded on the test data sheets.

4.5.3.2 Servicing and adjustments. The air conditioner shall be serviced and the brakes adjusted prior to the road test. Periodic servicing with lubricants and preventive maintenance adjustments shall be permitted during the testing at the intervals specified in the operation and service instruction handbook. Tires may be checked for proper inflation pressure. Other servicing, reconditioning, or repair shall not be permitted without prior approval of the procuring activity.

4.5.3.3 Service brakes. The air conditioner shall be loaded to design capacity. The air conditioner shall be subjected to three emergency stops in rapid succession from 20 mph on clean, dry, smooth, level, paved surface, free from loose material. Each stop shall be accomplished in not more than

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one-minute intervals. The air conditioner shall stop within 40 feet from the point the brakes are applied. A time-distance recorder shall be used and shall positively identify the point where the brakes are initially applied.

4.5.3.4 Parking brakes. The parking brakes shall cause those tires containing the brakes to skid and not roll when a towing force is applied to the air conditioner sufficient to make the vehicle move.

4.5.3.5 Ramp clearance. The air conditioner shall enter, negotiate, and leave a ramp with a slope of 17° in both forward and reverse directions. There shall be no contact between any part of the air conditioner, except the wheels, with any part of the ramp surface including the horizontal top and bottom landings. The length of the ramp, measured along the slope, shall be greater than the wheelbase of the air conditioner. The bottom and top of the ramp shall end in horizontal landings which form a sharply defined 17° angle with the ramp.

4.5.3.6 Towbar spring test. The towbar spring test shall consist of a cyclic dynamic part and a static part intended to demonstrate that the towbar spring complies with the requirements specified herein and will operate satisfactorily in normal use. The spring shall be subjected to 5000 cycles of operation which simulates raising and lowering the towbar to the full extremes. The spring shall also receive a static load test which simulates the effects of leaving the towbar extended over a long period of time. The static test shall be conducted over a 400 hour time period. After each 50 hours of test time, a spring force to angular displacement curve shall be prepared which can be used to establish the percentage loss of force. The spring shall be unacceptable if either part of the test causes the spring to be weakened such that it requires more than 25 pounds of force to lift the towbar from the fully extended to upright and locked position.

4.5.4 Electromagnetic interference test. The air conditioner, including chassis, shall comply with MIL-STD-461 as specified for class III C equipment with the exceptions specified herein. The air conditioner, including chassis, shall be tested in accordance with MIL-STD-462.

4.5.5 Acoustic noise level test. The sound levels produced by the air conditioner shall not exceed the values specified in table I at any of the measurement positions specified in 4.5.5.2. All measurements shall be made with the air conditioner operating at maximum rated capacity. The sound level meter shall be as specified in 4.5.5.3.

4.5.5.1 Background noise. Measurements shall be made only when the sound levels produced by the air conditioner are more than 10 decibels (dB) above the ambient background sound levels.

4.5.5.2 Measurement positions. Sound level measurements shall be made at 12 positions equally spaced every 30 degrees on 30- and 60-foot radius circles centered on the geometrical center of the air conditioner. All measurements shall be made with the air conditioner located on level, unobstructed pavement with the microphone 5 feet above the pavement level and oriented toward the air conditioner. The delivery hoses shall be attached to the outlet of the air conditioner and laid out to discharge the air away from the air conditioner. Sound level measurements shall also be taken at the operator's position.

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4.5.5.3 Instrumentation. The basic required instrumentation shall consist of a sound level meter having an A-weighted scale and conforming to the requirements specified in ANSI Standard S1.4. An up-to-date calibration (in dB ref: 20 $\mu N/m^2$) of the meter, including microphone, shall be available to permit correction of the measured data.

4.5.6 Tilted position test. The air conditioner shall be tilted in four different positions, 90° apart, at an angle of 8-1/2° to the true horizontal plane and checked for satisfactory operation of all components during tilt. Upon conclusion of this test, the air conditioner shall be examined for damage and checked for proper operation.

4.5.7 Leak test. A leak test shall be conducted on all components and refrigeration piping to determine the refrigerant leakage rate. A General Electric Type H-2 leak detector, or equal, shall be used to determine leaks. The combined total of all leaks shall not exceed 0.5 percent of the total refrigerant charge per year.

4.5.8 Operational tests. The air conditioner shall be subjected to and satisfactorily complete the following tests. Individual records of all operational tests shall be maintained.

4.5.8.1 The air conditioner shall be started and operated in the cooling mode at the prevailing ambient temperature for not less than 2 hours. The measured or indicated conditioned airflow rate shall be varied between 20 and 100 percent of rated airflow, and the conditioned air temperature shall not exceed 10.0°C (50°F) at a minimum static pressure of 35 inches wg.

4.5.8.2 The air conditioner shall also be started and operated in the heating and ventilating modes at the prevailing ambient temperatures for not less than 2 hours in each mode. The conditioned airflow rate shall be varied between 20 and 100 percent of rated airflow at pressures ranging between 0 and 35 inches wg static air pressure.

4.5.8.3 All operating and safety controls shall be checked to insure proper operation.

4.5.8.4 A functional test on all operating components, such as engine, compressor, brakes, switches, lights, doors, fans, blowers, and duct connectors shall be conducted to insure proper operation.

4.5.8.5 Dump valve test. The three 12-inch to 8-inch adapters shall each be connected to two 25-foot sections of flexible ducting which in turn is connected to the 12-inch outlets of the air conditioner. The dump value shall be opened while the air conditioner is operating against 20 inches wg static discharge pressure. The reactionary force shall not exceed the valve specified in paragraph 3.10.2.2. The dump valve and the conditioned airflow control dampers shall be fully opened while the blower motor and other components are monitored to assure that the air conditioner will operate satisfactorily under any of the ambient conditions specified herein without causing failure(s), safety devices to actuate, or unsafe operation of the air conditioner including dump valve discharge direction.

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4.5.9 Performance tests. The air conditioner shall be tested for 50 hours to demonstrate compliance with the cooling and heating capacities specified herein. Complete test data shall be recorded during capacity testing to demonstrate satisfactory performance of the air conditioner under the following conditions.

4.5.9.1 Basic rated cooling output test. The air conditioner shall be operated for 15 hours at an ambient condition of 37.8°C (100°F) DB and 29.5°C (85°F) WB. The air conditioner shall be operated at full cooling capacity at conditioned air delivery pressures ranging from 0 to 35 inches wg static air pressure. The airflow rate shall be at least 450 ppm and the conditioned air temperature shall not exceed 10.0°C (50°F).

4.5.9.2 Maximum rated cooling output test. The air conditioner shall be operated for 15 hours at an ambient condition of 51.7°C (125°F) DB and 23.9°C (75°F) WB. The air conditioner shall be operated at full cooling capacity at conditioned air pressures ranging from 0 to 35 inches wg static air pressure. The airflow rate shall be at least 450 ppm and the conditioned air temperature shall not exceed 10.0°C (50°F).

4.5.9.3 Basic rated heating output test. The air conditioner shall be operated for 15 hours at full heating capacity at the prevailing ambient conditions. Airflow rates and conditioned air temperatures shall be varied over the full range provided by the operating controls.

4.5.9.4 Ventilating output test. The air conditioner shall be operated for 5 hours in the ventilation mode at the prevailing ambient conditions. The air conditioner shall be operated over the full range of airflow rates and delivery pressures available.

4.5.10 Electrical tests. The air conditioner shall be given thorough electrical tests to determine that all circuits are inherently sound so that the requirements of this specification will be obtained. The electrical tests shall include, but not necessarily be restricted to, the following tests:

4.5.10.1 Continuity tests. Each electrical component and each cord and cable shall be given a continuity test to ascertain if it is wired and connected correctly and that good electrical contact is obtained.

4.5.10.2 Operating voltage tests. The operating voltage at all important points shall be checked for conformance to those shown on the circuit label and schematic drawing. This shall be accomplished with all controls set for normal operation of the air conditioner.

4.5.11 Duct storage test. The six 25-foot lengths of air ducts and the three air duct adapters shall be stored internally in the air conditioner. The air conditioner shall be operated in each mode of operation while the ducts and duct adapters are stored in the storage racks provided. No damage to air conditioner components or to the ducts and duct adapters shall occur as a result of this test.

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4.5.12 Environmental tests. The tests specified in table II shall be conducted in accordance with the specified methods and procedures of MIL-STD-810. Unless otherwise specified, operation of the air conditioner shall be required during these tests.

4.5.12.1 High temperature test. The highest operating temperature shall be as specified in 3.7.1.

4.5.12.2 Low temperature test. The lowest operating temperature shall be as specified in 3.7.1.

4.5.12.3 Shock test. The air conditioner is not required to operate during the shock test.

TABLE II. Environmental tests.

Test Method No.	Method Title	Procedures
501.1	High temperature	I
502.1	Low temperature	I
505.1	Solar radiation (sunshine)	I
506.1	Rain	I
507.1	Humidity	V
508.1	Fungus	I
509.1	Salt fog	I
510.1	Dust (Fine Sand)	I
516.2	Shock	II

4.5.13 Endurance test. A 900-hour endurance test shall be conducted to demonstrate compliance with the 435-hour MTBF specified in 3.5.1. The endurance testing shall be performed according to the following time schedule.

OPERATING REQUIREMENTS

Time (Hours)	Mode	Performance condition
150	Cooling	See 3.7.2
150	Cooling	See 3.7.3
200	Cooling	See 3.7.4
200	Heating	See 3.7.6, 3.7.6.1, and 3.7.7
200	Ventilating	Varied Ambient -53.9° to 51.7°C (-65° to 125°F)

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During the endurance testing, airflow and static pressure delivery capability shall be the specified maximum values. Engine speed shall be that required to provide the cooling output specified in 3.7.2 or 3.7.3, whichever is greater.

4.5.14. Maintainability demonstration. The air conditioner shall demonstrate the following maintainability features while causing minimal disturbance to other air conditioner components.

a. Removal of the diesel engine and refrigerant compressor without interference of structural members of other air conditioner components.

b. Drive belts shall be easily replaced and shall be provided with quick-access doors for visual inspection.

c. Evaporator and condenser coils shall be located so interference with structural members or other components does not hinder their removal.

d. The air conditioner cabinet and frame assembly shall be independent of the trailer chassis in order that a new trailer chassis could be fastened to the air conditioner enclosure and frame assembly.

e. Electric motors shall be mounted so that they can be easily removed and are accessible to maintenance personnel.

f. Operating controls and instrumentation shall be located so that all components are readily accessible by maintenance personnel using a minimum number of tools.

g. Condenser fans and conditioned air blowers shall be located for easy removal and replacement.

h. Sight gages, fluid level indicators, and filter pressure drop indicators shall be located behind quick-access doors or panels so they can be visually inspected without entering the air conditioner enclosure.

4.5.15 Final inspection. The air conditioner operating controls, electrical controls, refrigeration components, heat exchangers, trailer chassis and frame enclosure, fans and blowers, and engine drive shall be visually inspected to the maximum practical extent. Indication of physical breakdown or defects in any of these components shall be cause for rejection of the items.

4.6 Inspection of preservation and packaging and packing. The inspection of the preservation and packaging and packing shall be in accordance with the instructions in Section 5 or the specifications referenced therein.

5. PACKAGING

5.1 Preservation, packaging, and packing of chassis and engine. Preservation, packaging, and packing of the chassis and engine shall be to the appropriate level (see 6.2) in accordance with MIL-STD-281, mobile condition.

5.2 Preservation, packaging, and packing of air conditioners. Preservation, packaging, and packing of air conditioners shall be to the appropriate level (see 6.2) in accordance with MIL-R-3593.

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5.3 Shipment marking. In addition to any special marking required by the contract (see 6.2), interior packages and exterior container marking shall be in accordance with MIL-STD-129. The nomenclature shall be as follows:

Air Conditioner, A/M 32C-18,
Trailer Mounted, Diesel Engine Driven

6. NOTES

6.1 Intended use. The A/M 32C-18 air conditioner is intended for use in conditioning cargo, electronic, and personnel compartments during ground servicing and checkout of various aircraft.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification

b. Location and conditions for product conformance testing (see 4.4)

c. Selection of applicable levels of preservation, packaging, and packing (see Section 5)

d. Whether the air conditioners are to be shipped with required fluids and lubricants (see 3.13.5)

e. Special shipment marking (see 5.3)

f. Data and delivery requirements shall be as specified on the Contract Data Requirement List, (DD 1423).

6.3 Definitions. For the purpose of this specification, the following definitions will apply:

6.3.1 Air conditioner. The air conditioner is defined as including all components which make up the complete trailer mounted unit.

6.3.2 Reliability. Reliability is defined as the probability of performing a specific function under given conditions without failure for a specified period of time. Recognizing that, in general, the rate of failure of equipment is fairly constant throughout the life of equipment, the probability of nonfailure over an operating time interval decreases exponentially as a function of the length of the interval, during which time there is a constant failure rate, and can be expressed as follows:

$$\text{MTBF} = \frac{\text{Total test time}}{2.3}$$

$$\text{Reliability} = e^{-x} \text{ where } x = \frac{\text{Mission time}}{\text{MTBF}}$$

The $2.3 = \frac{4.61}{2}$ is based on the constant for the Poisson/Chi squared distribution, assuming an exponential (2 degree freedom) distribution even through the failure rate for the test time specified is to be zero.

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6.3.2.1 Mean-time-between-failures. The mean-time-between-failures is the arithmetic mean (average) of the operating time between failures.

6.3.3 Mission time. Mission time (t) is the expected or average operating time for each use of the item and for this specification is defined as 6 hours.

6.3.4 Exposed items. Exposed items are defined as exterior components and all components and parts that are exposed to the circulating condenser air.

6.3.5 Highway. A highway is defined as a road of level concrete or macadam.

6.3.6 Unimproved road. An unimproved road is defined as a road consisting of level-to-rolling gravel trails.

6.3.7 Weatherproof. So constructed or protected that exposure to the weather will not interfere with successful operation of the unit.

6.3.8 Guarded. Covered, shielded, enclosed, or otherwise protected by means of suitable covers, casings, barriers, screens, mats, or plaforms to remove the likelihood of approach or contact by persons of objects to a point of danger.

6.3.9 Continuous duty. Operation at a substantially constant load for an indefinitely long period of time.

6.3.10 Automatic. Self-acting, operating by its own mechanism when actuated by some impersonal influence, as for example, a change in current strength, pressure, temperature, or mechanical configuration.

6.4 Abbreviations. In order to save space and to avoid the spelling out of repetitious words and phrases, the following abbreviations have been used in the text of this specification:

ac	alternating current
amp	ampere
ANSI	American National Standards Institute
ASHRAE	American Society of Heating, Refrigeration, and Air Conditioning Engineering, Inc.
ASME	American Society of Mechanical Engineers
BTU/hr	British thermal units per hour
dB	decibel
DB	dry bulb
dBμV/M	decibel microvolt per meter
DC	direct current

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fpm	feet per minute
g	gravitational unit
Hg	mercury
KHz	kilohertz
MHz	megahertz
mph	miles per hour
MTBF	mean-time-between-failure
NFPA	National Fire Protection Association
ppm	pounds per minute
psig	pounds per square inch gage
RH	Relative Humidity
V	volt
WB	wet bulb
wg	water gage
$\mu N/m^2$	micro newtons per meter squared

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